

P2933R4

Extend `<bit>` header function with overloads for `std::simd`

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Abstract

Proposal to extend `std::simd` with overloads from other C++ standard libraries

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§ 1. Motivation

[P1928R7] introduced data parallel types to C++. It mostly provided operators which worked on or with `std::simd` types, but it also included overloads of useful functions from other parts of C++ (e.g., `sin`, `cos`, `abs`). In this paper we propose some other functions from standard C++ headers which should receive overloads to work with `std::simd` types. The list isn't exhaustive, but reflects those functions which are desirable to include.

§ 2. Support for `<bit>`

The `<bit>` header is part of the numerics library and provides utilities for manipulating and querying the properties of integral values when treated as collections of bits. The table below summarises the contents of `<bit>`.

Name	Purpose	Proposed (Y/N)
<code>endian</code>	A type which indicates the endianness of scalar types.	N
<code>bit_cast</code>	reinterpret the object representation of one type as that of another	N
<code>byteswap</code>	reverses the bytes in the given integer value	Y
<code>has_single_bit</code>	checks if a number is an integral power of two	Y
<code>bit_ceil</code>	finds the smallest integral power of two not less than the given value	Y
<code>bit_floor</code>	finds the largest integral power of two not greater than the given value	Y
<code>bit_width</code>	finds the smallest number of bits needed to represent the given value	Y
<code>rotr</code>	computes the result of bitwise left-rotation	Y
<code>rotr</code>	computes the result of bitwise right-rotation	Y
<code>countl_zero</code>	counts the number of consecutive 0 bits, starting from the most significant bit	Y

<code>countl_one</code>	counts the number of consecutive 1 bits, starting from the most significant bit	Y
<code>countr_zero</code>	counts the number of consecutive 0 bits, starting from the least significant bit	Y
<code>countr_one</code>	counts the number of consecutive 1 bits, starting from the least significant bit	Y
<code>popcount</code>	counts the number of 1 bits in an unsigned integer	Y

Of these types and functions, only the first two shouldn't be handled by `std::simd`:

- `endian` indicates the endianness of a scalar type. A SIMD value with elements of the underlying scalar type will have the same properties and does not do a special handling in `std::simd`.
- `bit_cast` should be handled differently for `std::simd` values, and a separate proposal for `simd_bit_cast` will be provided.

All the other functions from `<bit>` should be handled in `std::simd` by element-wise application of the function to each element of the SIMD value. Any constraints and behaviours on the function will be applied at the SIMD value level. For instance, if `byteswap` participates in overload resolution only if the argument type satisfies `std::integral` concept then the overload of `byteswap` with `std::simd` parameter has the same constraint for `std::simd<T, N>::value_type`.

One small modification to the behaviour of `<bit>` for `std::simd` is where the return type differs to the input type. For example, the standard `<bit>` header defines some query functions as returning integer values:

```
template< class T >
constexpr int bit_width( T x ) noexcept;

template< class T >
constexpr int countl_one( T x ) noexcept;
```

If an `int` were to be returned from the `std::simd` overload of such functions then the size of the elements could change. For example, computing the bit width of a 8-bit integer could generate a `std::simd` of 64-bit integers as the output, which would lead to a dramatic change in storage size and performance. Instead, we propose that all the overloads for `<bit>` should return element types which are the same physical size as the element types they are querying. This would mean that calling `bit_width` on an `unsigned` 8-bit integer will return a `std::simd` containing signed 8-bit values.

When calling the rotate functions `rotl` and `rotr` functions it is common to want to rotate all `simd` elements by the same amount. An overload will be provided which takes a scalar `int` value to match

`rotl` and `rotr` in the `<bit>` header. In this case there is no need to supply an integer of the same width as the first parameter's elements (as described above for the `simd` variant) since broadcasting an `integer` to a `simd` has negligible performance impact.

§ 3. Wording

Below, substitute the  character with a number the editor finds appropriate for the table, paragraph, section or sub-section.

§ 3.1. Modify `[version.syn]`

In `[version.syn]` bump the `__cpp_lib_simd` version.

§ 3.2. Modify `[simd.expos]`

NOTE: `simd-type` is also added by P2663R7 and it is exactly the same.

```
template<class V>
    concept simd-type = // exposition only
        same_as<V, basic_simd<typename V::value_type, typename V::abi_type>>
        is_default_constructible_v<V>;
```

```
template<class V>
    concept simd-floating-point = // exposition only
        same_as<V, basic_simd<typename V::value_type, typename V::abi_type>>
        is_default_constructible_v<V> && floating_point<typename V::value_ty
```

§ 3.3. Update the synopsis

In the header `<simd>` synopsis - [simd.syn] - add at the end after the "Mathematical functions"

```
// [simd.bit], Bit manipulation
template<simd-type V> constexpr V byteswap(const V& v) noexcept;
template<simd-type V> constexpr V bit_ceil(const V& v) noexcept;
template<simd-type V> constexpr V bit_floor(const V& v) noexcept;

template<simd-type V>
    constexpr typename V::mask_type has_single_bit(const V& v) noexcept;

template<simd-type V0, simd-type V1>
    constexpr V0 rotl(const V0& v, const V1& s) noexcept;
template<simd-type V>
    constexpr V rotl(const V& v, int s) noexcept;

template<simd-type V0, simd-type V1>
    constexpr V0 rotr(const V0& v, const V1& s) noexcept;
template<simd-type V>
    constexpr V rotr(const V& v, int s) noexcept;

template<simd-type V>
    constexpr rebind_simd_t<make_signed_t<typename V::value_type>, V>
        bit_width(const V& v) noexcept;
template<simd-type V>
    constexpr rebind_simd_t<make_signed_t<typename V::value_type>, V>
        countl_zero(const V& v) noexcept;
template<simd-type V>
    constexpr rebind_simd_t<make_signed_t<typename V::value_type>, V>
        countl_one(const V& v) noexcept;
template<simd-type V>
    constexpr rebind_simd_t<make_signed_t<typename V::value_type>, V>
        countr_zero(const V& v) noexcept;
template<simd-type V>
    constexpr rebind_simd_t<make_signed_t<typename V::value_type>, V>
        countr_one(const V& v) noexcept;
template<simd-type V>
    constexpr rebind_simd_t<make_signed_t<typename V::value_type>, V>
        popcount(const V& v) noexcept;
```

§ 3.4. Add new section `[simd.bit]` after `[simd.math]`

◆ `basic_simd` bit library `[simd.bit]`

```
template<simd-type V> constexpr V byteswap(const V& v) noexcept;
```

Constraints:

The type `V::value_type` models `integral`.

Returns:

A `basic_simd` object where the i^{th} element is initialized to the result of `std::byteswap(v[i])` for all i in the range `[0, V::size())`.

```
template<simd-type V> constexpr V bit_ceil(const V& v) noexcept;
```

Constraints:

The type `V::value_type` is an unsigned integer type (`[basic.fundamental]`).

Preconditions:

For every i in the range `[0, V::size())`, the smallest power of 2 greater than or equal to `v[i]` is representable as a value of type `V::value_type`.

Returns:

A `basic_simd` object where the i^{th} element is initialized to the result of `std::bit_ceil(v[i])` for all i in the range `[0, V::size())`.

Remarks: A function call expression that violates the precondition in the *Preconditions:* element is not a core constant expression (`[expr.const]`).

```
template<simd-type V> constexpr V bit_floor(const V& v) noexcept;
```

Constraints:

The type `V::value_type` is an unsigned integer type (`[basic.fundamental]`).

Returns:

A `basic_simd` object where the i^{th} element is initialized to the result of `std::bit_floor(v[i])` for all i in the range `[0, V::size())`.

```
template<simd-type V> constexpr typename V::mask_type has_single_bit(const
```

Constraints:

The type `V::value_type` is an unsigned integer type ([basic.fundamental]).

Returns:

A `basic_simd_mask` object where the i^{th} element is initialized to the result of `std::has_single_bit(v[i])` for all i in the range `[0, V::size())`.

```
template<simd-type V0, simd-type V1> constexpr V0 rotl(const V0& v0, const  
template<simd-type V0, simd-type V1> constexpr V0 rotr(const V0& v0, const
```

Constraints:

- The type `V0::value_type` is an unsigned integer type ([basic.fundamental]),
- the type `V1::value_type` models integral,
- `V0::size() == V1::size()` is true, and
- `sizeof(typename V0::value_type) == sizeof(typename V1::value_type)` is true.

Returns:

A `basic_simd` object where the i^{th} element is initialized to the result of `bit_func(v0[i], static_cast<int>(v1[i]))` for all i in the range `[0, V0::size())`, where `bit_func` is the corresponding scalar function from `<bit>`.

```
template<simd-type V> constexpr V rotl(const V& v, int s) noexcept;  
template<simd-type V> constexpr V rotr(const V& v, int s) noexcept;
```

Constraints:

The type `V::value_type` is an unsigned integer type ([basic.fundamental])

Returns:

A `basic_simd` object where the i^{th} element is initialized to the result of `bit-func(v[i], s)` for all i in the range `[0, V::size())`, where `bit-func` is the corresponding scalar function from `<bit>`.

```
template<simd-type V>
    constexpr rebind_simd_t<make_signed_t<typename V::value_type>, V>
        bit_width(const V& v) noexcept;
template<simd-type V>
    constexpr rebind_simd_t<make_signed_t<typename V::value_type>, V>
        countl_zero(const V& v) noexcept;
template<simd-type V>
    constexpr rebind_simd_t<make_signed_t<typename V::value_type>, V>
        countl_one(const V& v) noexcept;
template<simd-type V>
    constexpr rebind_simd_t<make_signed_t<typename V::value_type>, V>
        countr_zero(const V& v) noexcept;
template<simd-type V>
    constexpr rebind_simd_t<make_signed_t<typename V::value_type>, V>
        countr_one(const V& v) noexcept;
template<simd-type V>
    constexpr rebind_simd_t<make_signed_t<typename V::value_type>, V>
        popcount(const V& v) noexcept;
```

Constraints:

The type `V::value_type` is an unsigned integer type ([basic.fundamental])

Returns:

A `basic_simd` object where the i^{th} element is initialized to the result of `bit-func(v[i])` for all i in the range `[0, V::size())`, where `bit-func` is the corresponding scalar function from `<bit>`.

§ 4. Revision History

R3 => R4

- Minor typo fixes.
- Strike some unnecessary notes.

- LEWG confirmed that overloads for `rotl/rotr` should be provided which take a plain `int` as their second parameter.
- Modified constraints to be more simply worded.
- Modified returns test to match [\[P1928R15\]](#).

R2 => R3

- Add missing synopsis wording addition.
- Change detailed behaviour descriptions to match the style/wording of recent P1928 updates.
- Added concepts to detect simd types whose elements are integral, unsigned, and so on.
- Added overload to `rotl/rotr` to allow the second parameter to be a scalar.

R1 => R2

- Added feature test macro

R0 => R1

- Fix typo: `std::make_signed` to `std::make_signed_t`
- Make several paper text improvements

§ References

§ Informative References

[P1928R15]

Matthias Kretz. *std::simd - Merge data-parallel types from the Parallelism TS 2*. 22 November 2024. URL: <https://wg21.link/p1928r15>

[P1928R7]

Matthias Kretz. *std::simd - Merge data-parallel types from the Parallelism TS 2*. 15 October 2023. URL: <https://wg21.link/p1928r7>